

Generative AI — From Prediction to Creation

Dr Vaishali Kulkarni



Introduction

- ❑ What is Generative AI?
- ❑ How is it different from traditional AI and discriminative machine learning?
- ❑ How do generative models learn to create new content?
- ❑ Where are large generative models used in language and vision?

Introduction

- Connecting Prior Knowledge to Generative AI
- **Building on Previous Courses**
- **Machine Learning**
 - Introduced core concepts: supervised/unsupervised learning
 - Foundation for statistical algorithms and evaluation metrics
- **Deep Learning**
 - Basic foundation of neural network
 - Provided knowledge of: CNNs, RNNs
- **Natural Language Processing**
 - Covered language representation, tokenization, sequence modeling
 - Provided the basic foundation for text understanding and generation
- **Advanced Topics in ML**
 - Explored cutting-edge methods: meta learning, Graph Neural networks
 - Enabled powerful generative architectures like GANs, VAEs and Transformer

Introduction

Discriminative model	Generative model
Predicts a label or output	Produces a new sample or content
Learns boundaries between classes	Learns patterns or structure in data
“Is this email spam?”	“Generate a realistic email”
“Is this image a cat?”	“Generate an image of a cat”
“What sentiment does this review express?”	“Write a positive product review”

- A discriminative model learns to distinguish.
- A generative model learns enough about the data to create something similar but new.
- Discriminative modelling is often associated with learning $P(y|x)$.
- Generative modelling attempts to learn the data distribution $P(x)$, or the joint relationship $P(x, y)$.

Introduction

- **Why Study Generative AI?**
- **Innovates content creation:** text, images, music, code
- **Transforms industries:** healthcare, education, design, robotics
- **Merges learning & creativity:** from modeling distributions to generating novel outputs
- **Prepares you for frontier research & applications**

“Generative AI isn't just the next chapter in ML—it's a new language for creation.”

Introduction

Characteristic	Generative Adversarial Networks (GAN)	Variational Autoencoders (VAE)	Transformer-Based Models	Diffusion Models
 Core Mechanism	Two networks in competition	Encode and decode with randomness	Tokenization and self-attention	Add noise, reverse the process
 Content Generation	Creates fake content	Generates variations of input	Generates text like humans	Generates content from noise
 Key Feature	Adversarial training	Data compression	Self-attention	Reverse diffusion

Introduction

Comparison of Generative AI Models

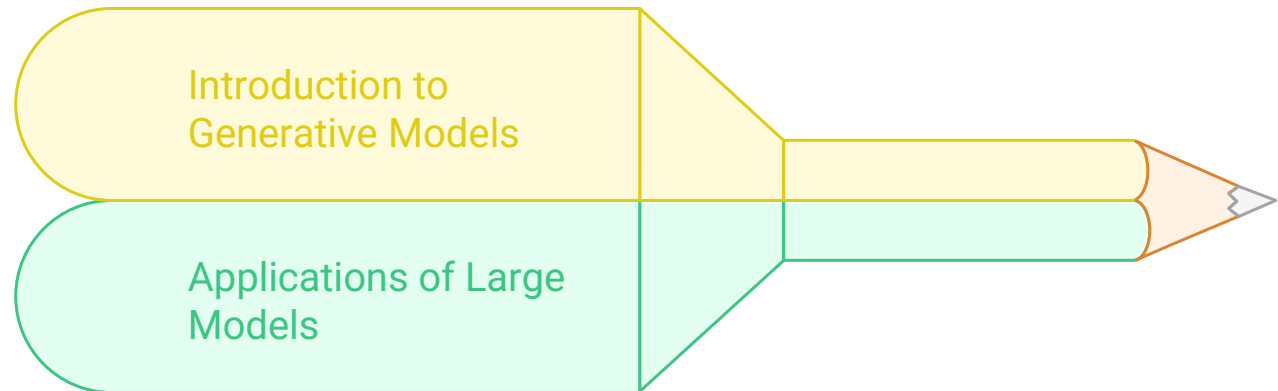
Characteristic	Flow-based Models	Autoregressive Models
Generation Process	Transforms random data through reversible steps	Generates content piece by piece sequentially
Reversibility	Each step is reversible	Not inherently reversible
Probability Calculation	Calculates exact likelihood of content	Likelihood calculation is complex
Example	Transforming noise into an image	ChatGPT generating text

Made with  Napkin

Introduction

- **Foundations of Generative AI**
- **Unit 1**
- Introduction to Generative Models,
- Applications of Large Models for Vision and Language.

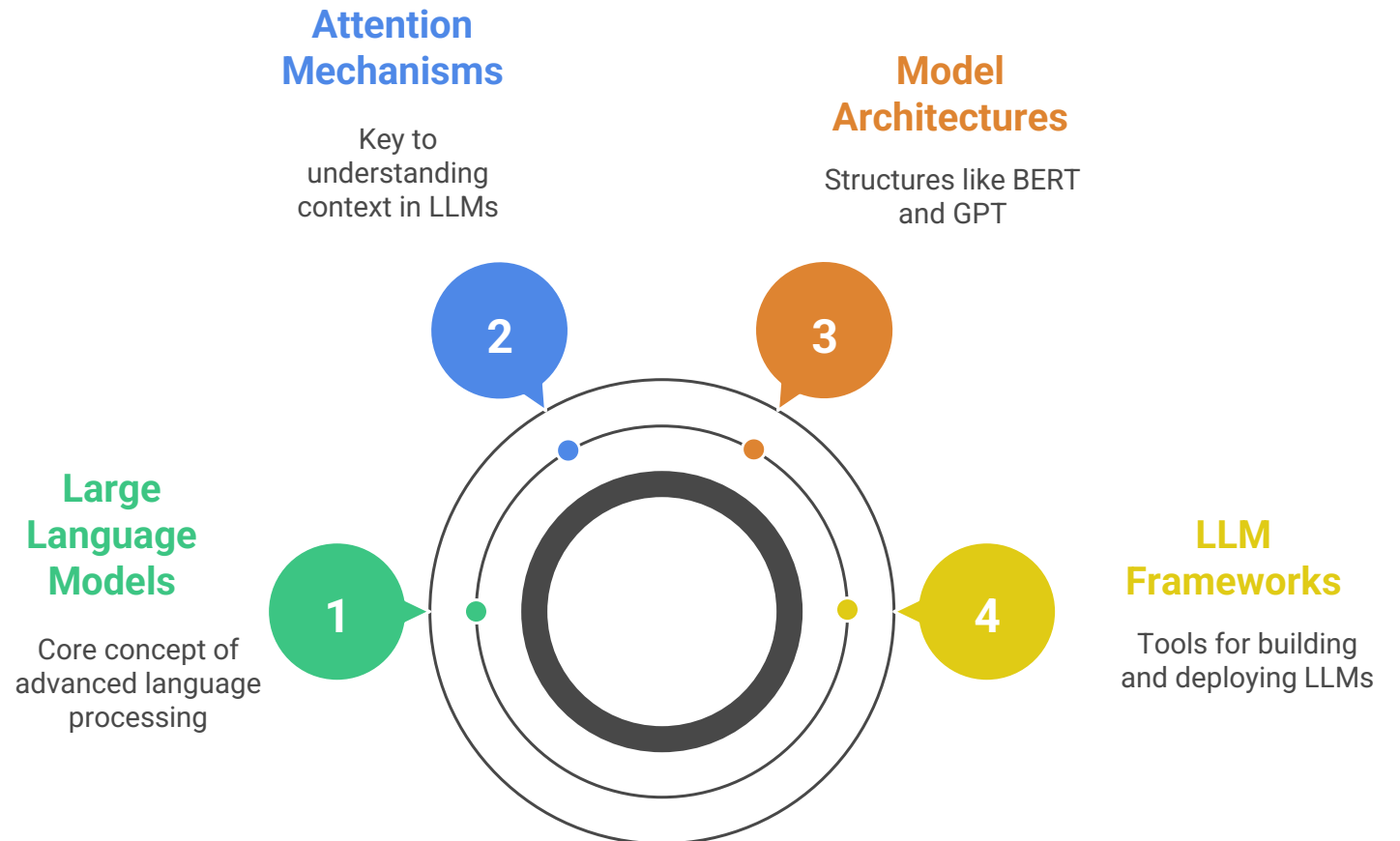
Building the Future of AI



Introduction

- Unit 2
- Large Language Models (LLM)
- Introduction to LLM
- Positional Embeddings(PE)
- Encoder-only model
- Decoder-only model
- Encoder-Decoder model
- LLM Frameworks

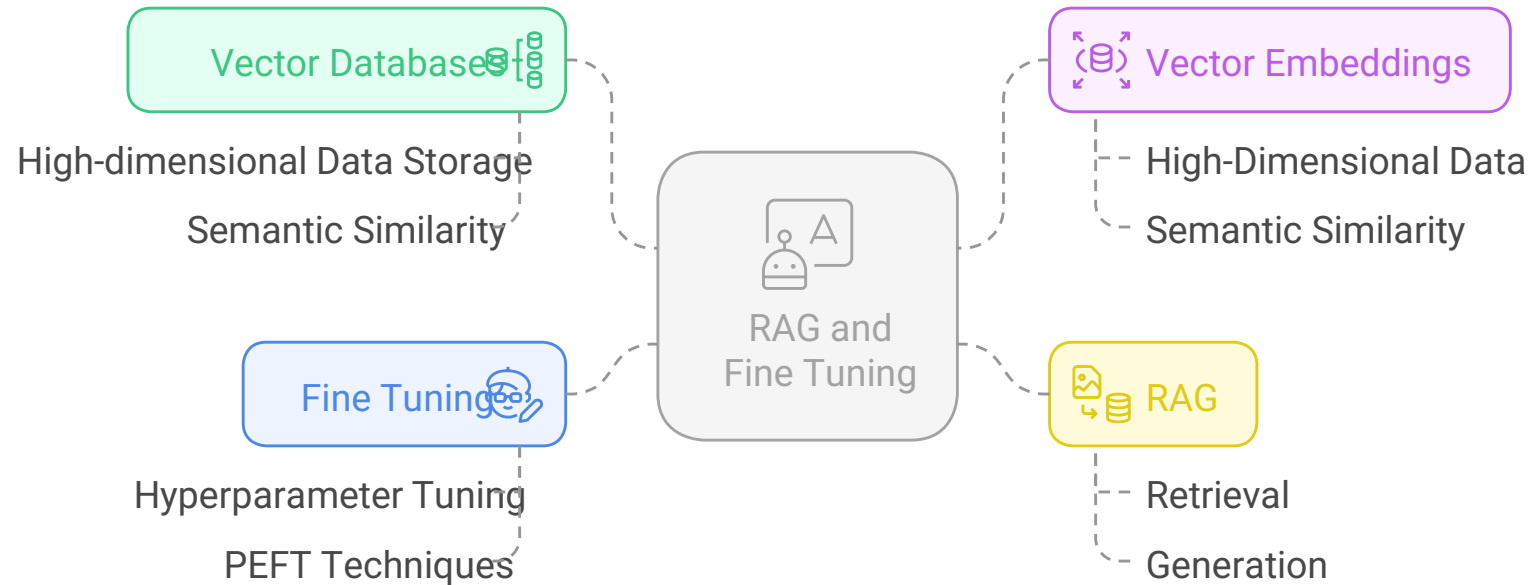
Large Language Models Architecture



Introduction

- **Unit 3**
- **RAG and Fine Tuning**
- Vector Databases
- Retrieval-Augmented Generation (RAG)
- Full fine Tuning v/s Parameter Efficient Fine Tuning (PEFT) Techniques, LoRA, QLoRA

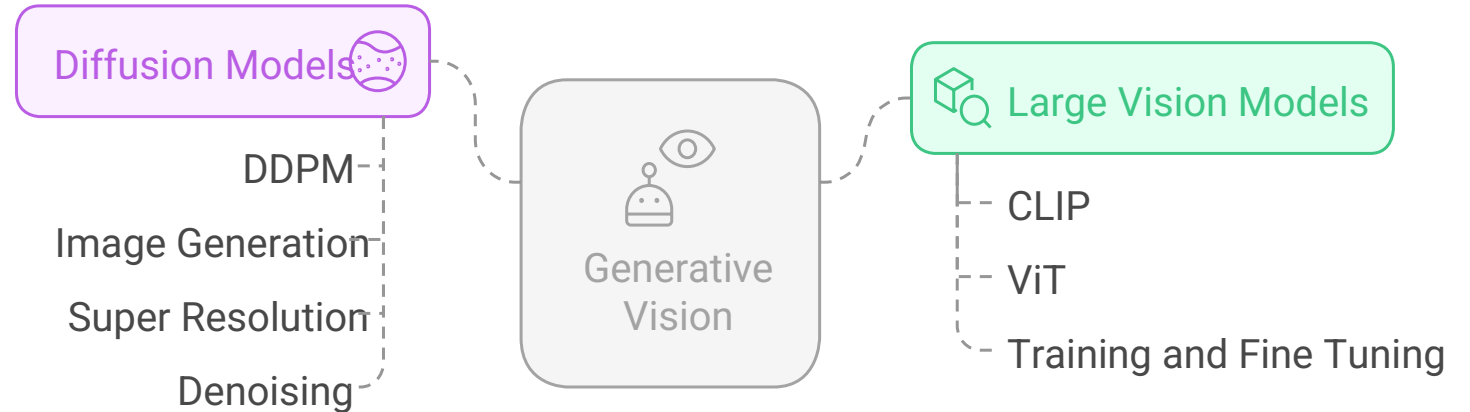
RAG and Fine Tuning Techniques



Introduction

- **Unit 4**
- **Generative Vision with Diffusion and Large Vision Models (LVM)**
- Overview of LVMs (CLIP, ViT), architectures, applications
- Introduction to Diffusion: Training DDPM for image generation, super resolution and denoising.

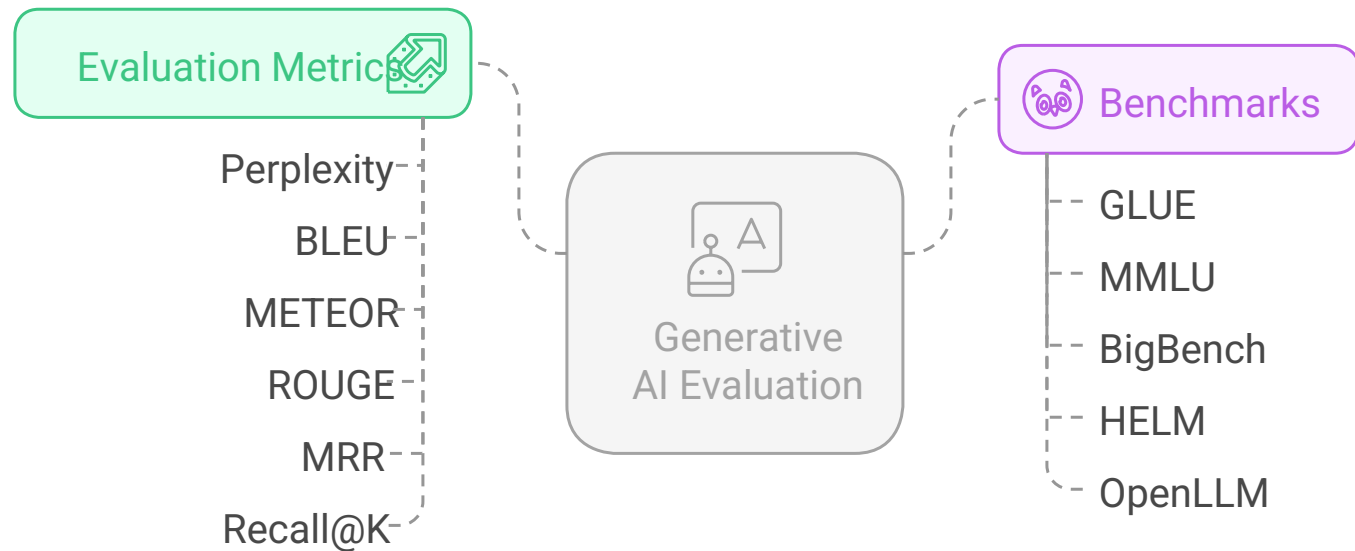
Generative Vision with Diffusion and LVMs



Introduction

- **Unit 5**
- **Generative AI Evaluation**
- Evaluation of LLM:
 - Perplexity (Language Modelling),
 - BLEU, METEOR, ROUGE
- Benchmarks:
 - GLUE, MMLU, BigBench, HELM, OpenLLM.

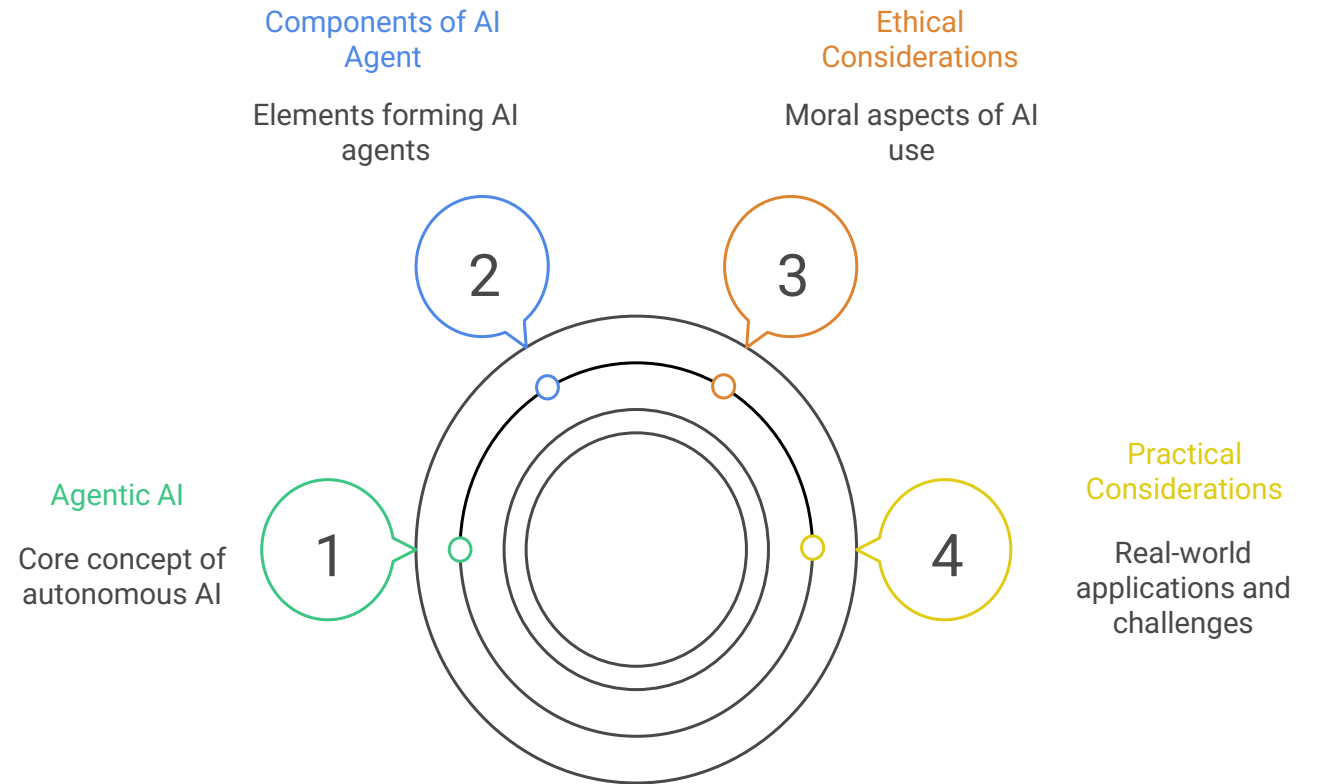
Generative AI Evaluation Metrics and Benchmarks



Introduction

- Unit 6
- Introduction to Agentic AI and its Applications

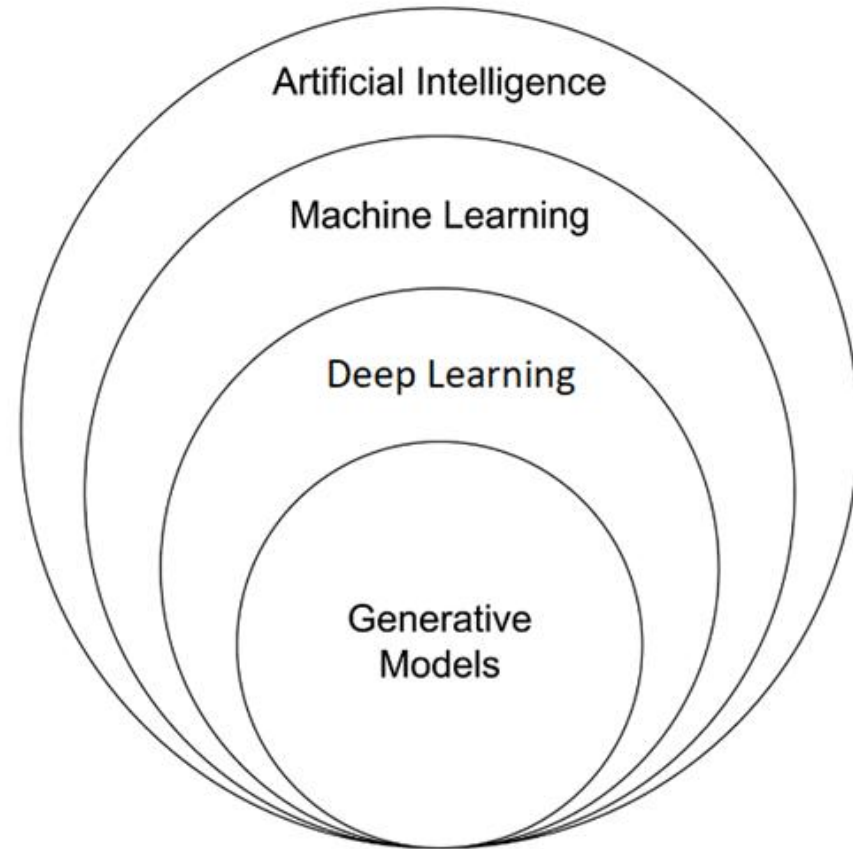
Agentic AI Structure



Introduction

Generative Artificial Intelligence (Gen AI)

Generative Artificial Intelligence (Gen AI) is a set of AI algorithms that utilize existing content like text, audio files, or images to create new content



Introduction

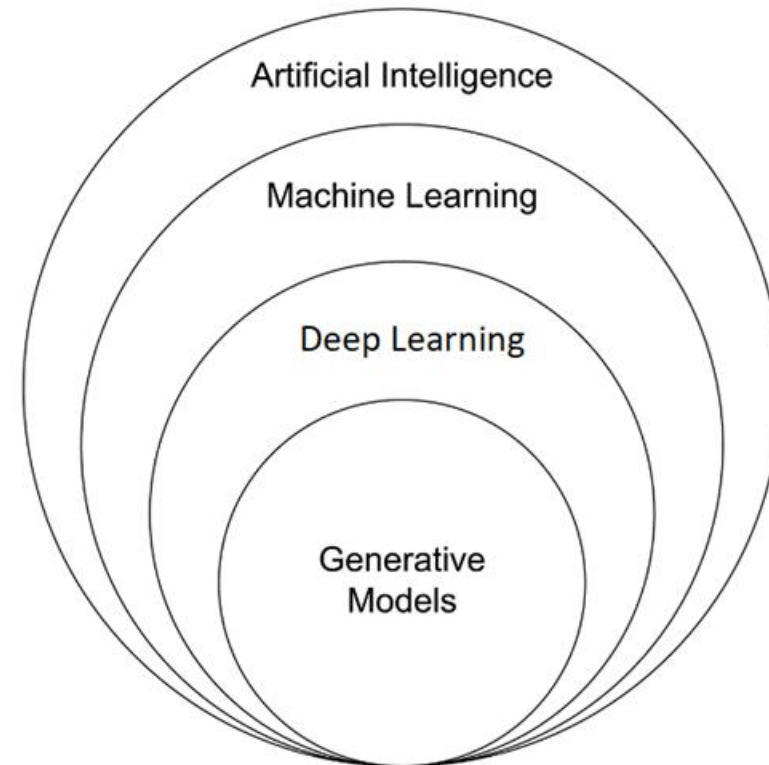
Generative Artificial Intelligence (Gen AI)

Key Characteristics:

1.Content Creation: Generates human-like text, realistic images, or other forms of media.

2.Versatility: Can be applied across industries, including education, entertainment, healthcare, and more.

3.Interactivity: Adapts to user prompts or contextual inputs to generate meaningful and personalized outputs.



Introduction

Generative Artificial Intelligence (Gen AI)

GenAI snapshot (2024–2025):

ChatGPT: ~700M weekly active users (Sep 2025).

Adoption: 71% of orgs report regular gen-AI use (McKinsey, 2025).

Spending: GenAI spending forecast \$644B in 2025 (+76% YoY) (Gartner).

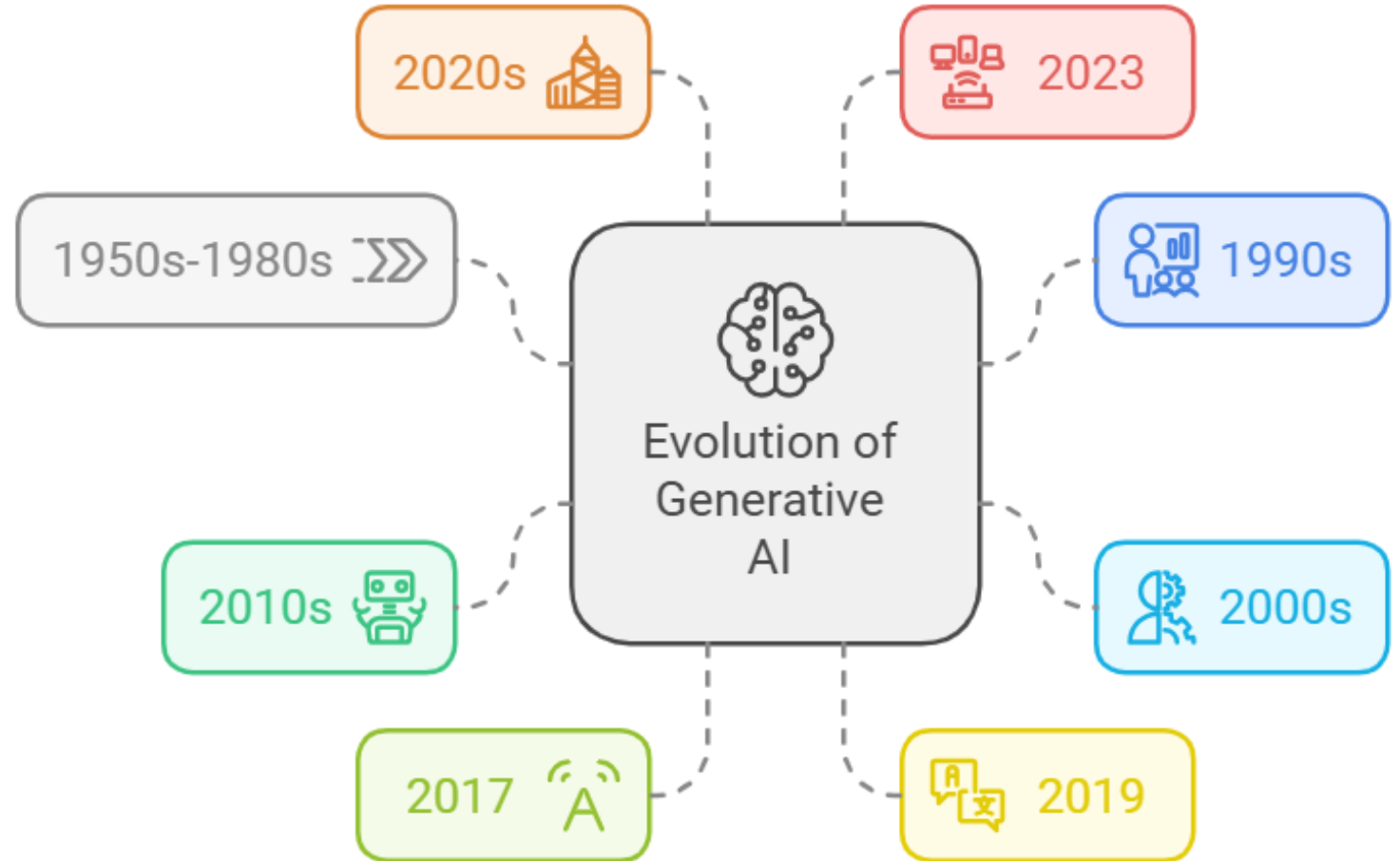
Investment: \$33.9B private investment in GenAI in 2024 (+18.7%) (Stanford AI Index).

Developer tools: GitHub Copilot >20M all-time users (Microsoft FY2025).

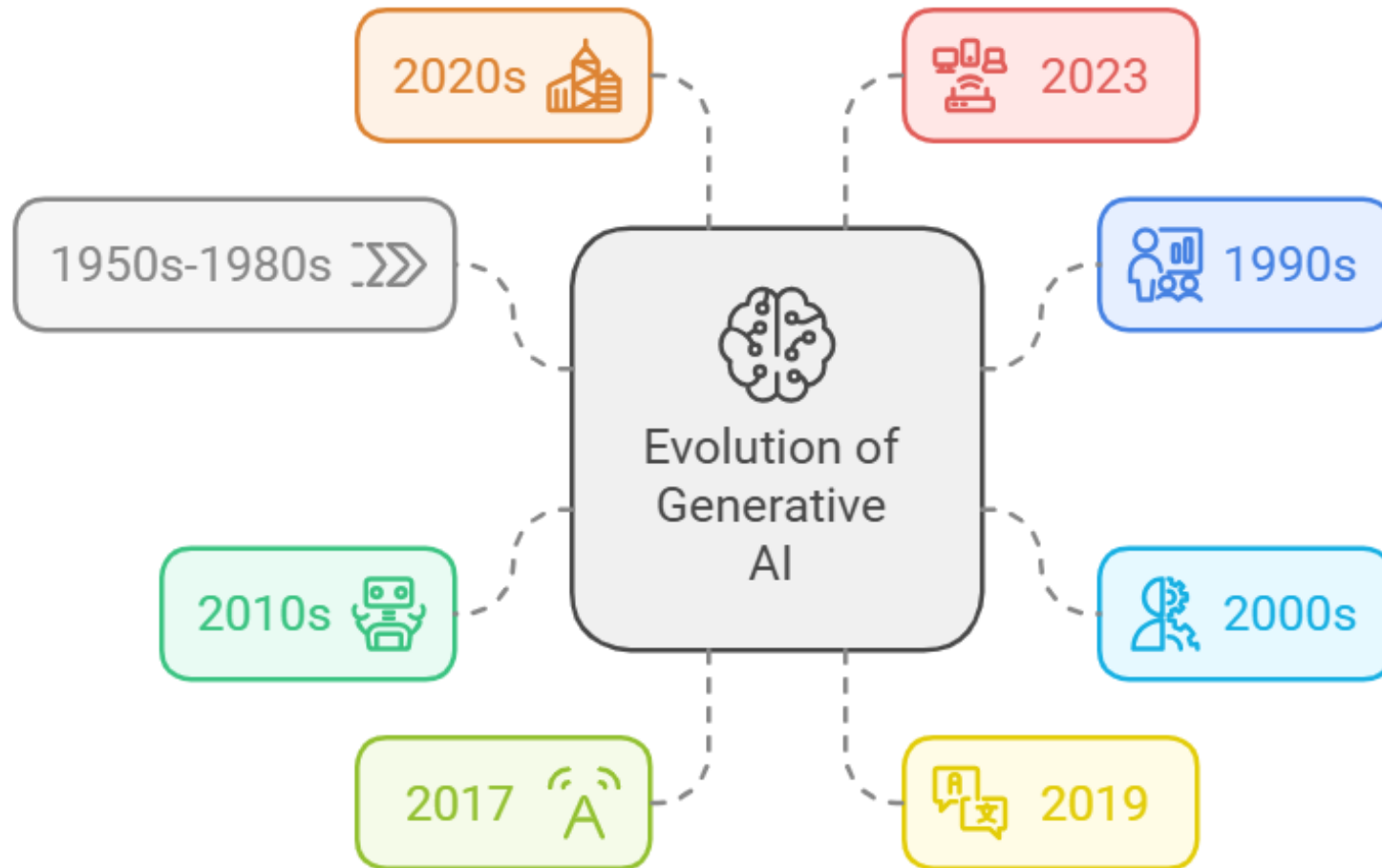
Evolution of Generative AI

1950s - 1980s: The Foundations

- **1956:** The term "Artificial Intelligence" is coined at the Dartmouth Conference, marking the beginning of AI as a field of study.
- **1960s:** Early explorations of machine learning and neural networks begin, laying the groundwork for future generative models.
- **1986:** Geoffrey Hinton and others publish a paper on backpropagation, a key algorithm for training neural networks.



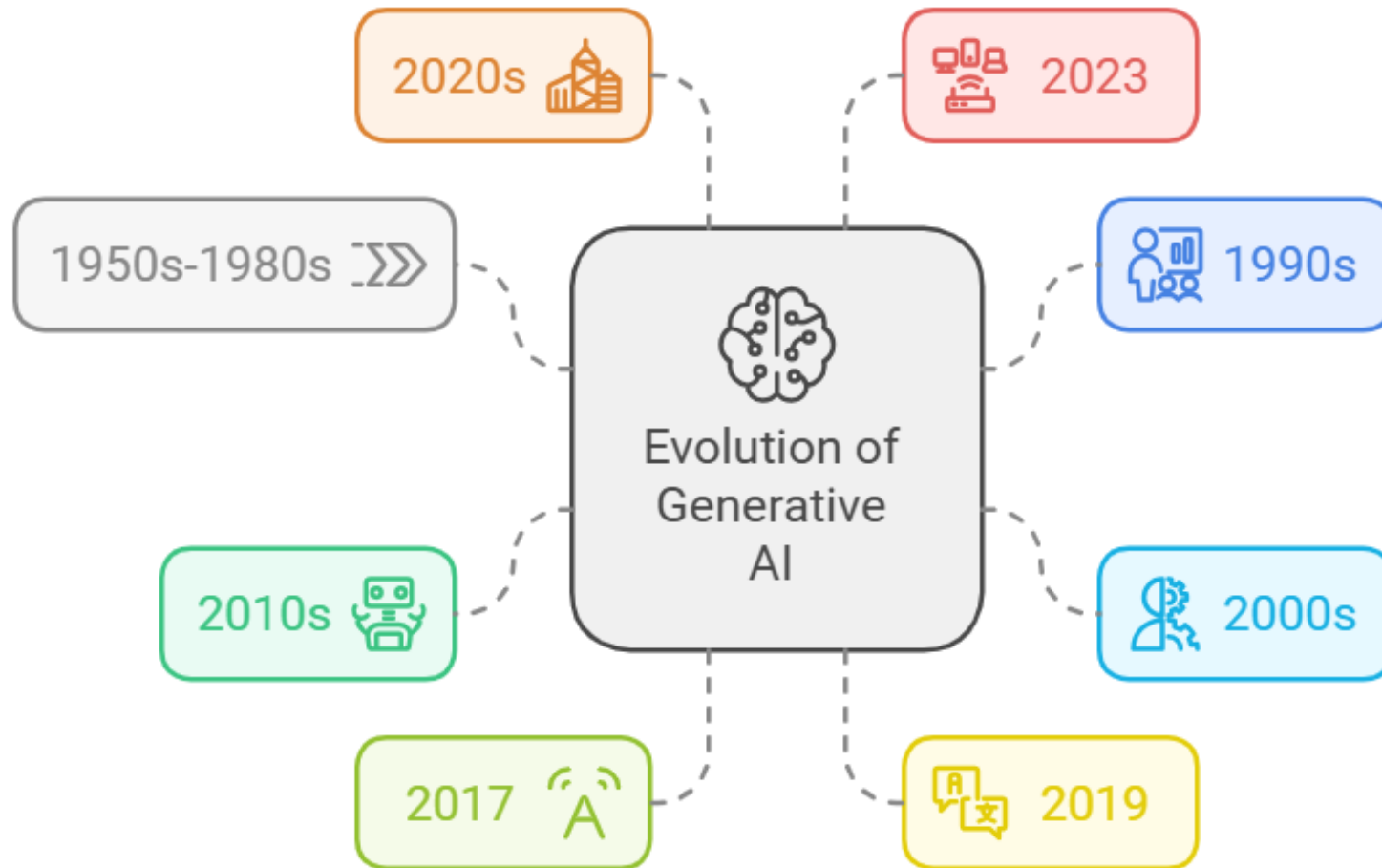
Evolution of Generative AI



1990s: The rise of statistical learning

1996: Hidden Markov Models (HMMs) become dominant in speech recognition.
1997: LSTMs are introduced, enabling better long-range sequence modeling.
Late 1990s: Probabilistic graphical models & EM-based methods become mainstream.

Evolution of Generative AI



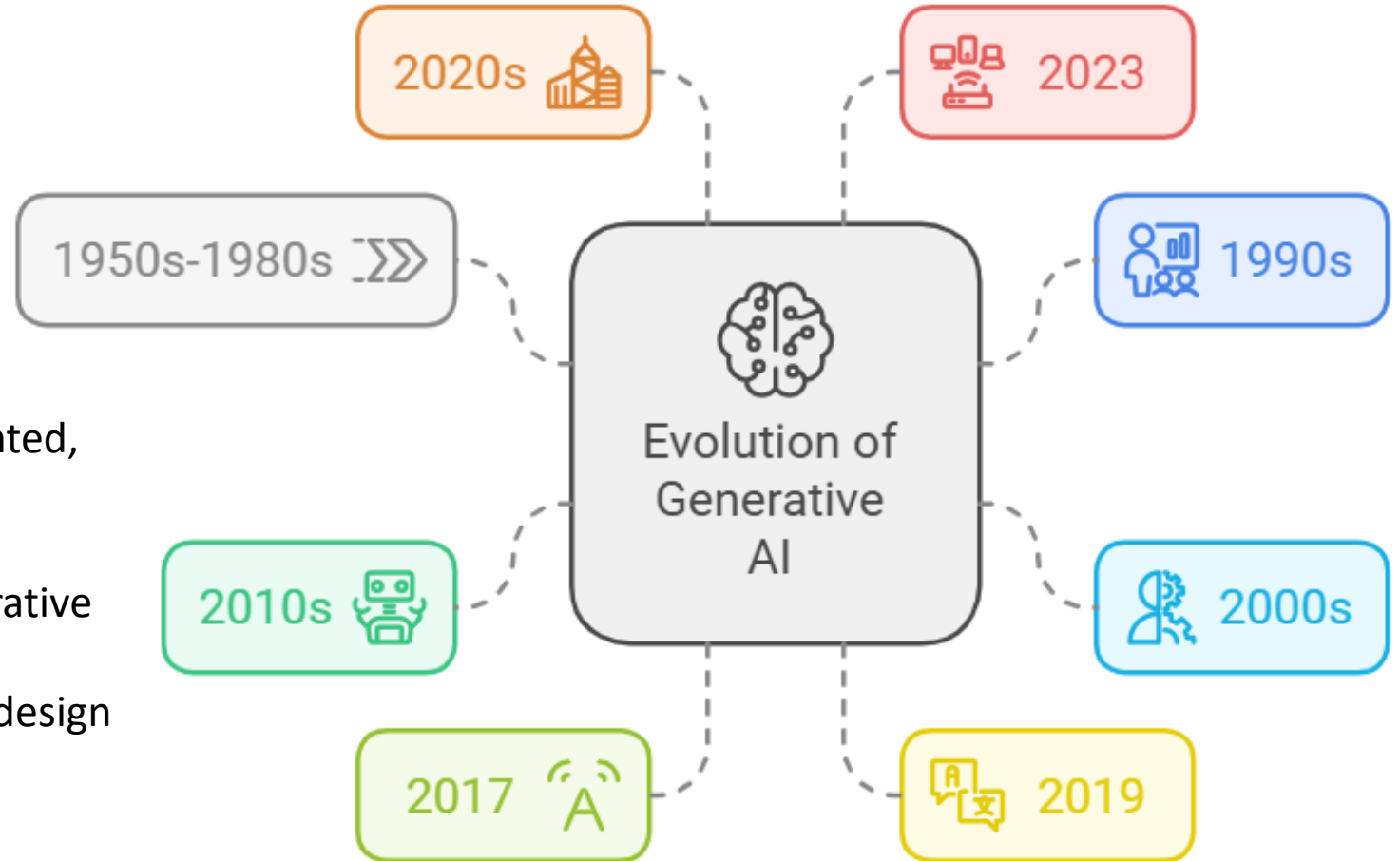
2000s: Probabilistic + deep generative foundations

2003: Latent Dirichlet Allocation (LDA) popularizes topic modeling.

2006: Deep Belief Networks revive deep learning via unsupervised pretraining.

Late 2000s: Larger datasets + GPUs set the stage for modern deep generative models.

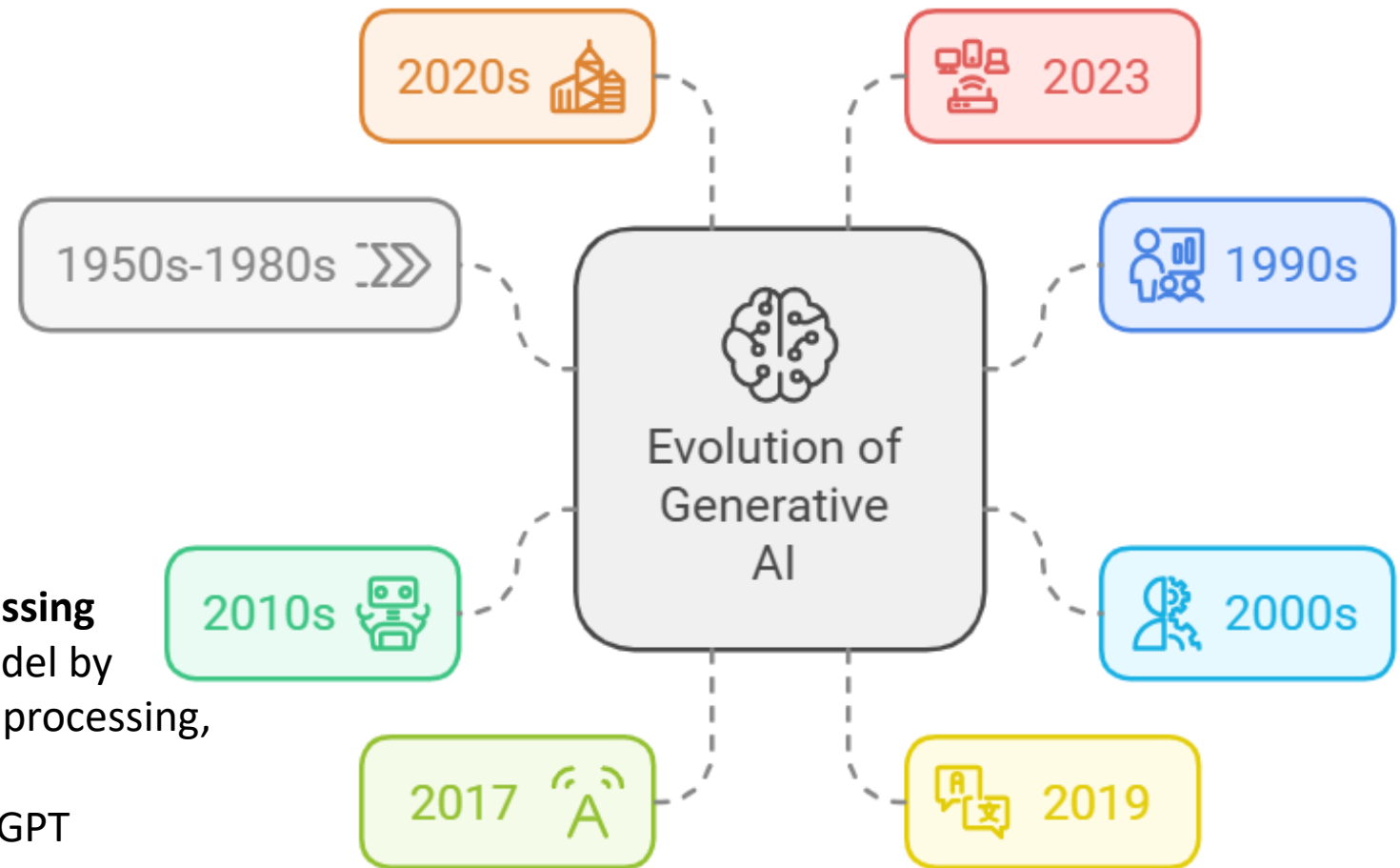
Evolution of Generative AI



2010s: Breakthroughs in Generative AI

- **2014:** The first GANs are successfully implemented, allowing for the generation of realistic images.
- **2015:** Variational Autoencoders (VAEs) are introduced, providing a new approach to generative modeling.
- **2016:** The first applications of GANs in art and design emerge, showcasing the creative potential of generative models.

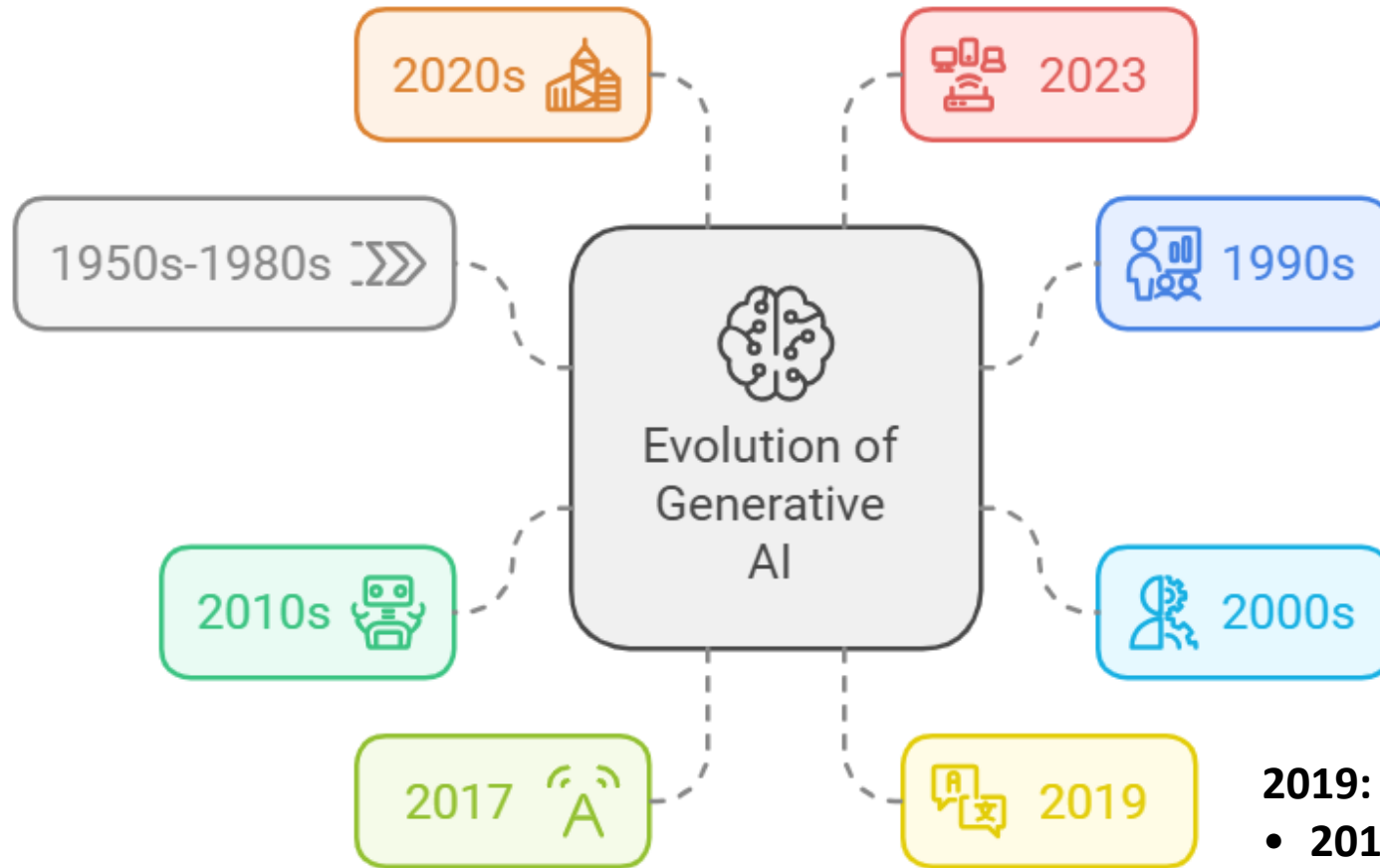
Evolution of Generative AI



2017: Advancements in Natural Language Processing

- **2017:** The introduction of the Transformer model by Vaswani et al. revolutionizes natural language processing, enabling more sophisticated text generation.
- **2018:** OpenAI releases the first version of the GPT (Generative Pre-trained Transformer) model, demonstrating impressive capabilities in text generation.

Evolution of Generative AI



2019: The Era of Large Language Models

- **2019**: OpenAI releases GPT-2, a significantly larger and more powerful language model that can generate coherent and contextually relevant text.
- **2019**: The introduction of BERT (Bidirectional Encoder Representations from Transformers) by Google enhances understanding in natural language processing tasks.

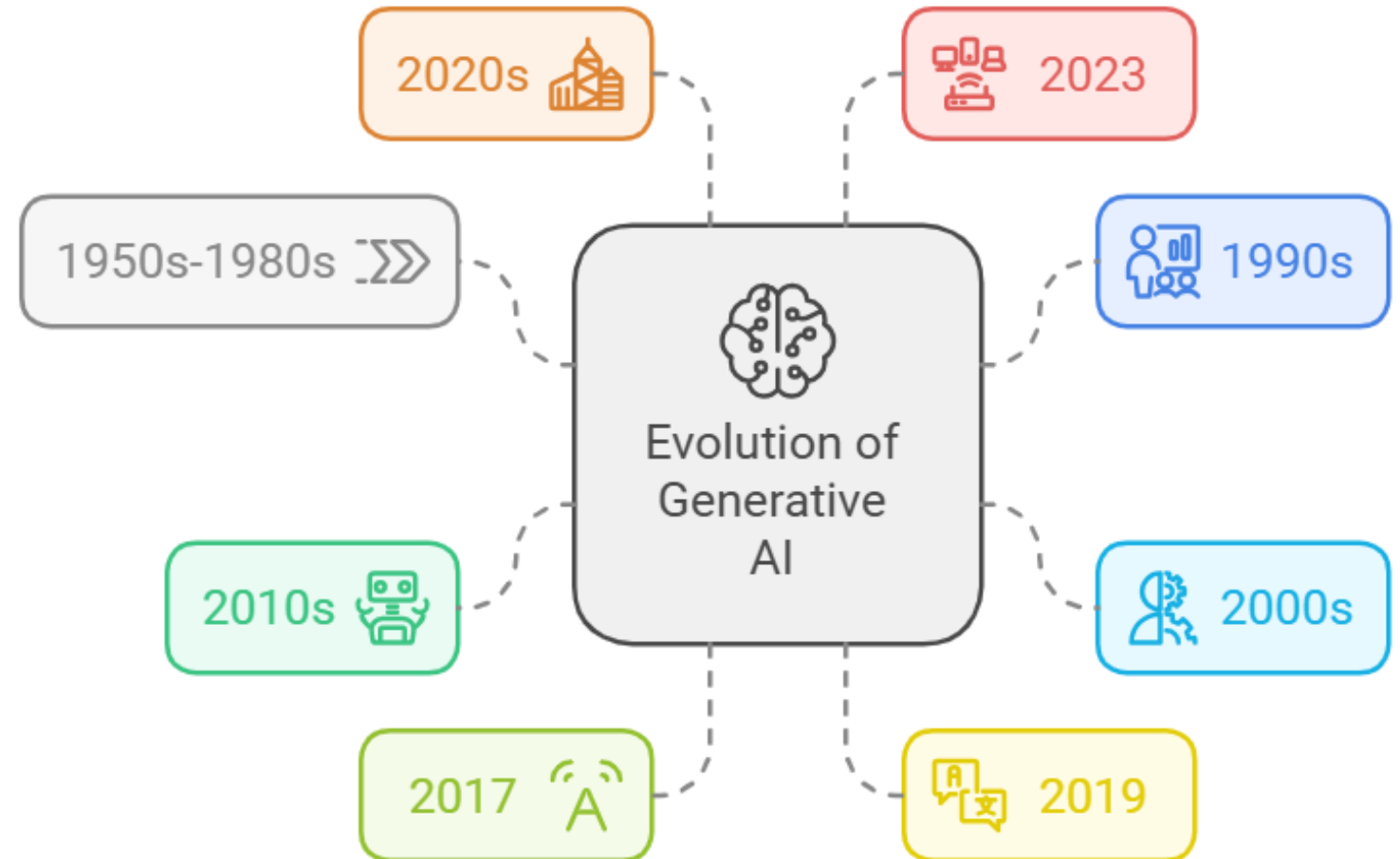
Evolution of Generative AI

2020s: Generative AI goes mainstream

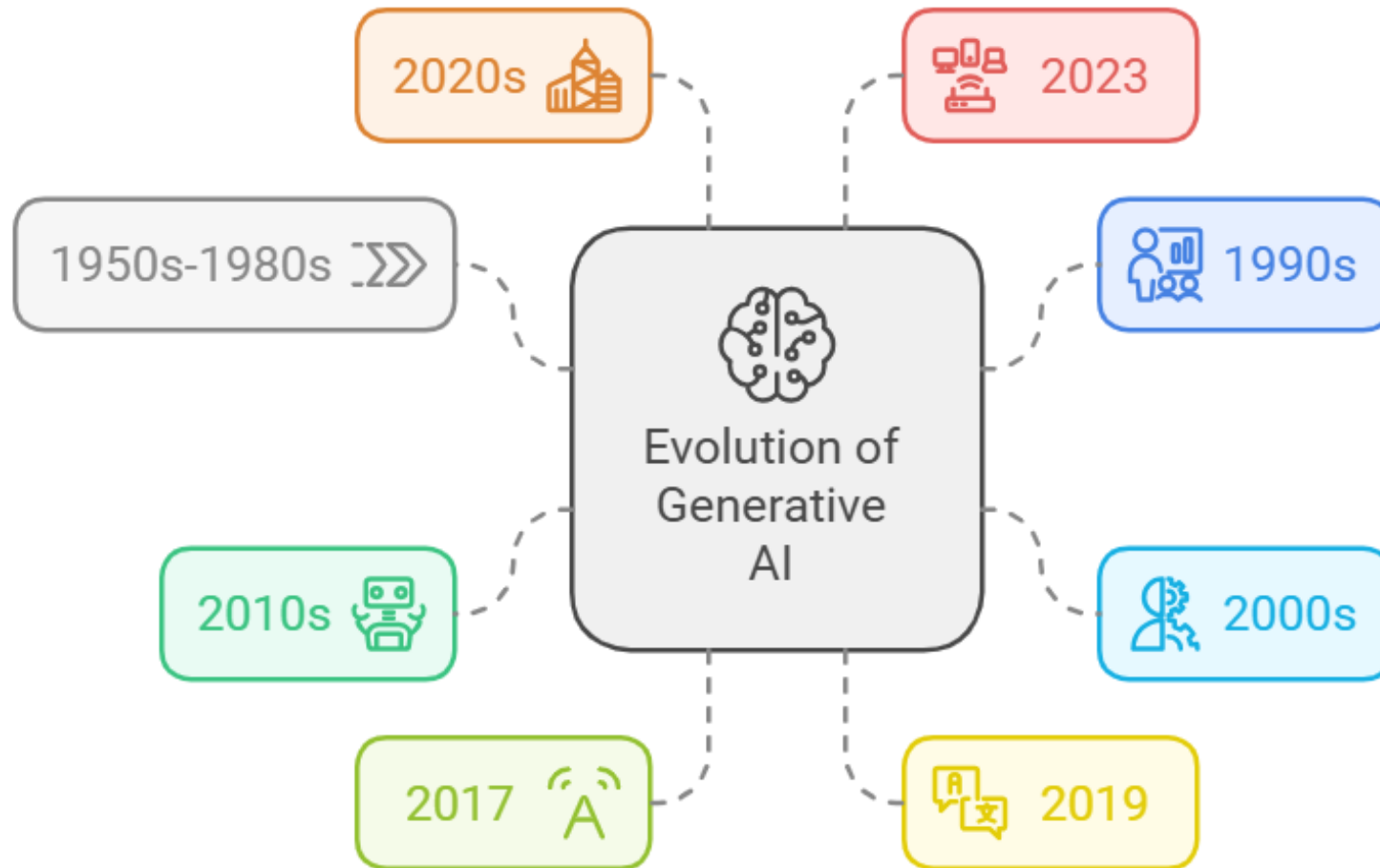
2020: GPT-3 scales LLMs (175B parameters).

2021–2022: Diffusion models + text-to-image tools accelerate (e.g., DALL·E, Stable Diffusion).

Late 2022: ChatGPT launches and rapidly mainstreams conversational AI.



Evolution of Generative AI



2023–2025: Multimodal + long-context era

2024: GPT-4o, Gemini 1.5, Claude 3 and Llama 3 expand multimodal & long-context capabilities.

2025: ChatGPT reaches ~700M weekly active users; enterprise/autonomous workflows accelerate.

Sources: OpenAI (GPT-4o, May 2024); Google (Gemini 1.5, Feb 2024); Anthropic (Claude 3, Mar 2024); Meta (Llama 3, Apr 2024); OpenAI (ChatGPT usage report, Sep 2025).

Introduction

Generative Artificial Intelligence

Market outlook (2025+):

AI spending forecast ~ \$1.5T in 2025 (Gartner).

Global AI market ~ \$235B in 2024 → > \$631B by 2028 (IDC).

Economic impact: GenAI could add \$2.6T–\$4.4T annually (McKinsey).

Long-run: GenAI market projected to reach ~\$1.3T by 2032
(Bloomberg Intelligence).

Evolution of Generative AI

...2026

- It's been three-and-a-half years since generative AI exploded onto the scene.
- In this past year, progress has continued its relentless pace:
- Vibe coding took off,
- Companies embraced agentic workflows,
- Regular users of ChatGPT hit 900 million and Google's Gemini surpassed 750 million,
- OpenAI posted an \$852 billion valuation in its latest funding round.

Evolution of Generative AI

...2026

- AI capability is not plateauing.
 - It is accelerating and reaching more people than ever.
 - Industry produced over 90% of notable frontier models in 2025, and several of those models now meet or exceed human baselines on PhD-level science questions, multimodal reasoning, and competition mathematics.
- The U.S.-China AI model performance gap has effectively closed.
- AI models can win a gold medal at the International Mathematical Olympiad but cannot reliably tell time—an example of what researchers call the jagged frontier of AI.
 - Gemini Deep Think earned a gold medal at IMO, yet the top model reads analog clocks correctly just 50.1% of the time.
 - AI agents made a leap from 12% to ~66% task success on OSWorld, which tests agents on real computer tasks across operating systems, though they still fail roughly 1 in 3 attempts on structured benchmarks.

Evolution of Generative AI

...2026

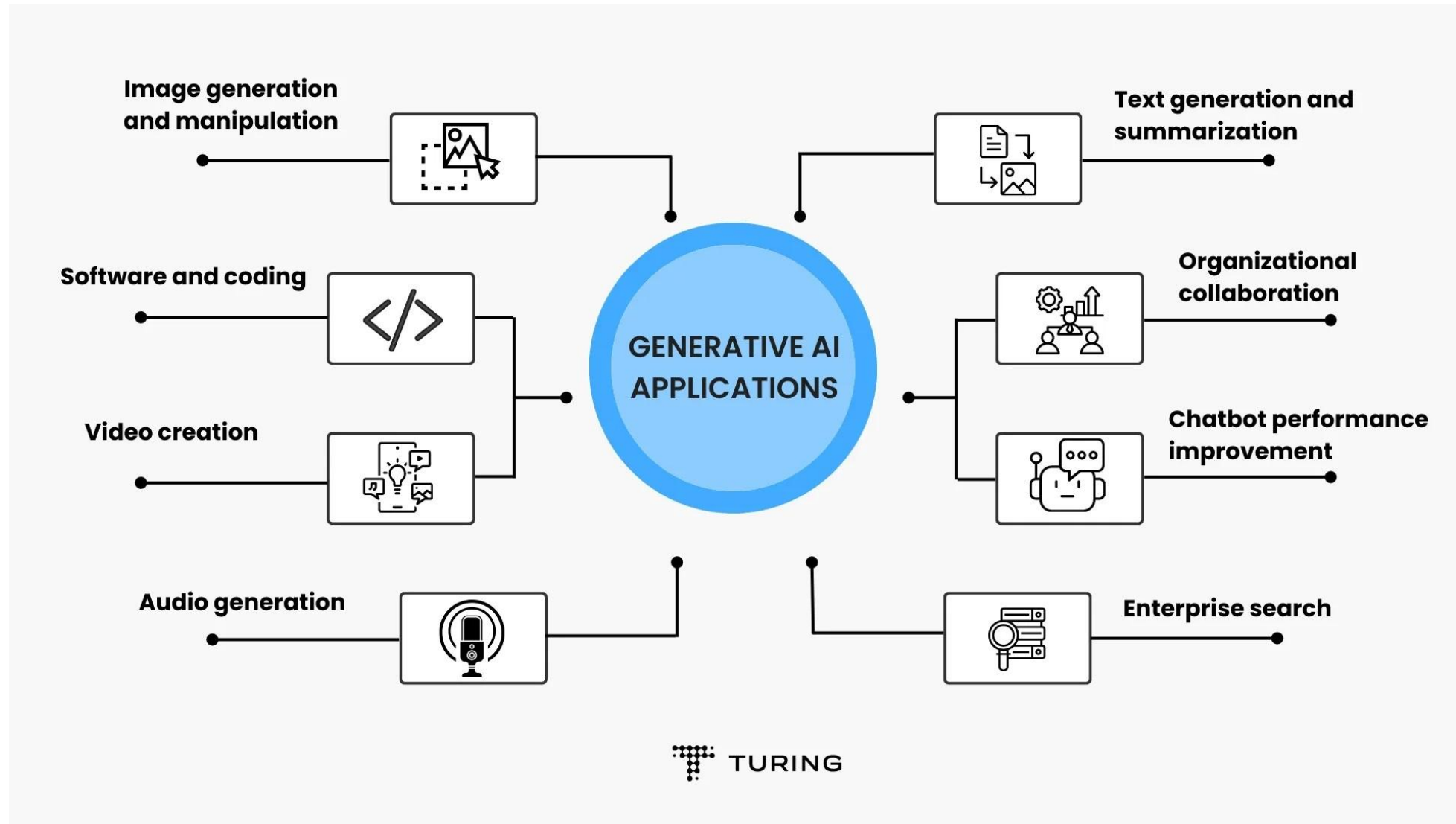
- AI adoption is spreading at historic speed, and consumers are deriving substantial value from tools they often access for free.
 - Generative AI reached 53% population adoption within three years, faster than the PC or the internet, though the pace varies by country and correlates strongly with GDP per capita.
- AI's environmental footprint is expanding alongside its capabilities. Grok 4's estimated training emissions reached 72,816 tons of CO2 equivalent.
 - AI data center power capacity rose to 29.6 GW, comparable to New York state at peak demand, and annual GPT-4o inference water use alone may exceed the drinking water needs of 1.2 million people.

Evolution of Generative AI

...2026

- Robots still fail at most household tasks, even as they excel in controlled environments.
 - Robots succeed in only 12% of household tasks, highlighting how far AI is from mastering the physical world.
 - On RLBench, robotic manipulation in software-based simulations has reached 89.4% success, but the gap between predictable lab settings and unpredictable household environments is wide.
- Responsible AI is not keeping pace with AI capability, with safety benchmarks lagging and incidents rising sharply.
 - Almost all leading frontier AI model developers report results on capability benchmarks, but reporting on responsible AI benchmarks remains spotty.
 - Documented AI incidents rose to 362, up from 233 in 2024.
 - Adding to the challenge, recent research found that improving one responsible AI dimension, such as safety, can degrade another, such as accuracy.

Applications of Generative AI



Applications of Generative AI

GenAI Content Generation Types

Characteristic	Description	Examples
Text Generation	Produces text explaining concepts	ChatGPT, Google Gemini, Perplexity AI
Image Generation	Creates, alters, and produces images	DALL-E 2, Midjourney, Stable Diffusion
Music/Audio Generation	Creates audio data and music	Aiva AI, Soundful
Video Generation	Creates videos, animations, and effects	Canva, DeepAI, Invideo
Code Generation	Creates or autocompletes code	Google's Gemini, Vertex AI, Sonar

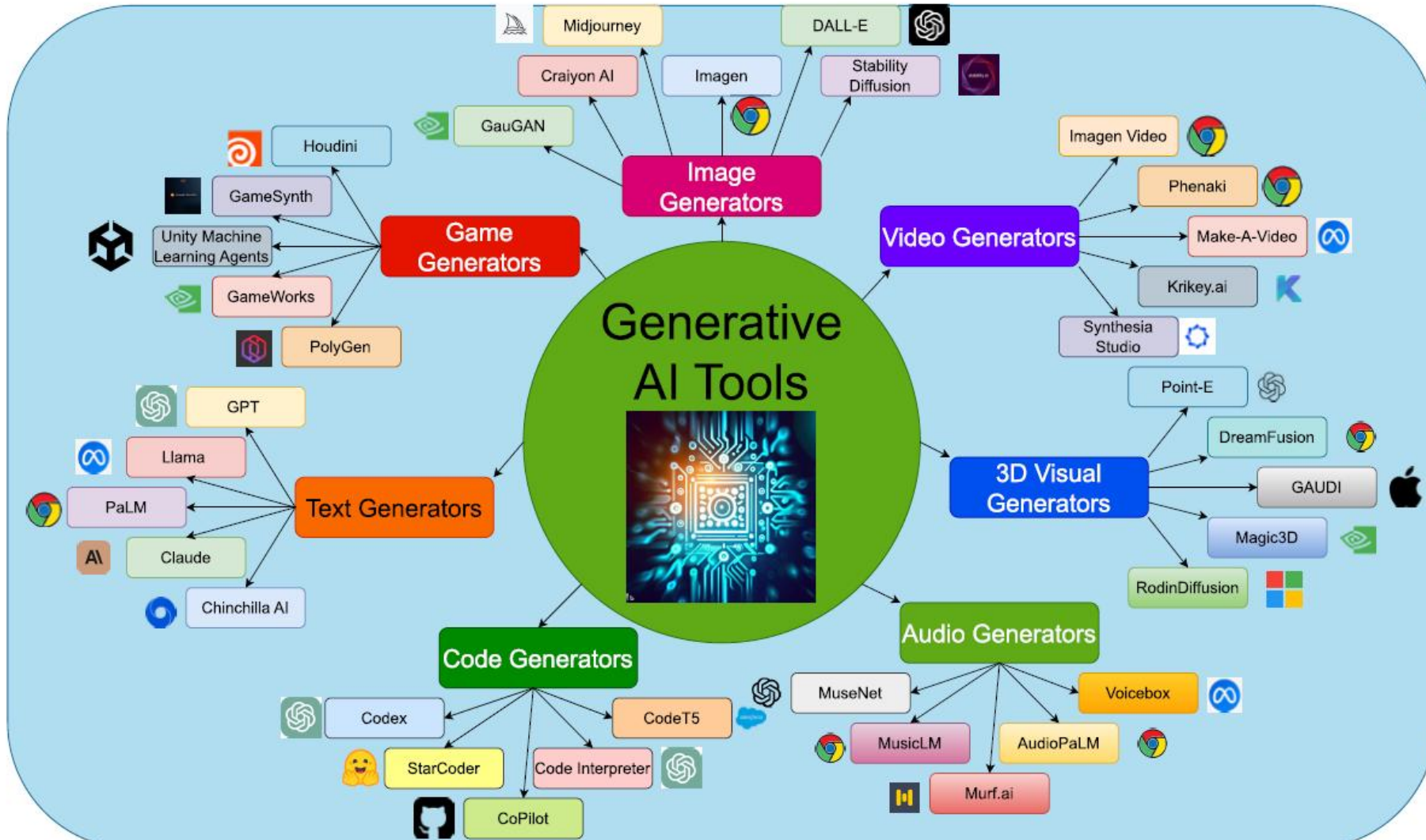
Applications of Generative AI

20 Surprising Functions And Fields Gen AI Is Revolutionizing

legal_documentation
product_development outbound_sales
digital_advertising disaster_response
financial_services_and_fintech manufacturing
pharmaceuticals_development space_exploration
sports_and_fitness_training linguistics
agriculture home_repair entertainment
energy_and_utilities_management
investment retail_pricing_and_promotions
banking
mathematics_research

“AI is the new electricity and it will transform every industry.”

Categories of Gen AI products

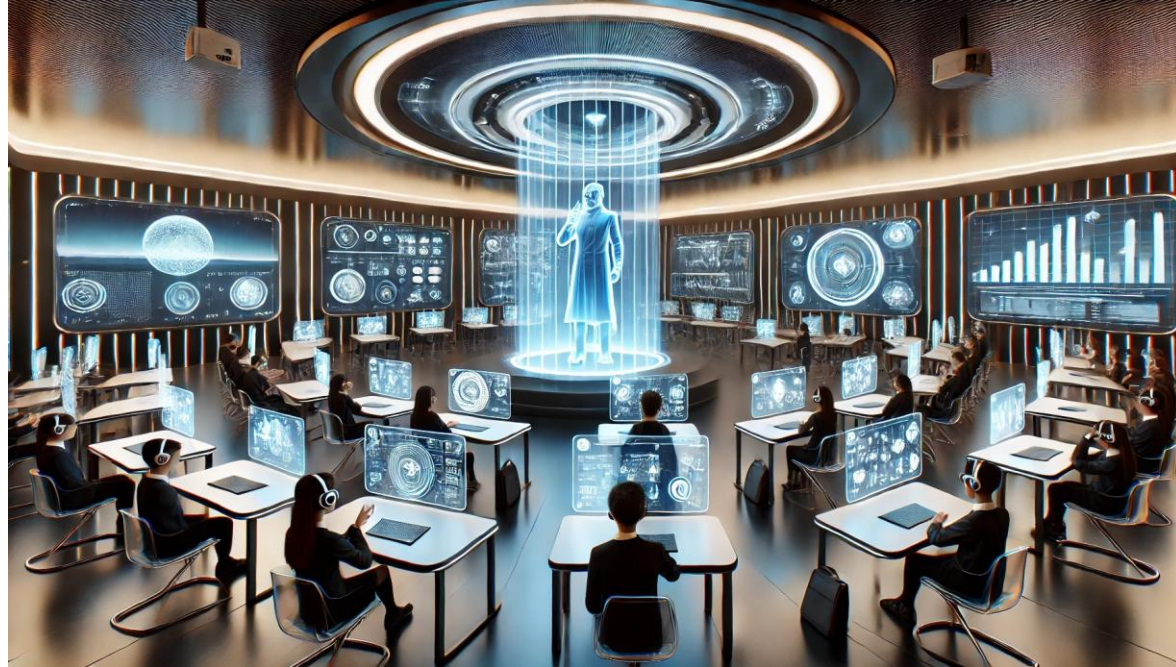


Some interesting examples....

Prompt used

Generate futuristic classroom with technology enabled teacher

Suggestions by the LLM after generating the image



The image has been created based on your request. To improve or modify it, you can choose one of the following ideas:

- Add more interactive elements, such as robotic assistants or AI-powered whiteboards, to further emphasize the futuristic theme.
- Incorporate natural elements like digitalized plants or a virtual outdoor scene projected on the walls for a more eco-friendly classroom aesthetic.
- Adjust the lighting to include dynamic effects, such as color transitions or star-like patterns on the ceiling, to enhance the immersive atmosphere.

Some interesting examples....

ChatGPT response



Introduction to Biometrics (PDFDrive ...
PDF

Prompt used

False Non-Match Rate

The **False Non-Match Rate (FNMR)** is a key performance measure for biometric systems. It indicates the expected probability that two samples of the same biometric trait, obtained from the same user, will incorrectly be identified as a non-match. This type of error often arises due to factors such as variability in user interaction with biometric sensors, environmental conditions, or the intrinsic limitations of biometric feature extraction algorithms.

Key Characteristics:

- **Definition:** FNMR refers to the proportion of mate samples (samples from the same person) that are falsely rejected by the system.
- **Impact:** A higher FNMR means more genuine users may face rejection during authentication, leading to inconvenience.
- **Mitigation:** Many systems address this issue by allowing repeated attempts or by improving sensor accuracy to reduce intra-user variability .

Some interesting examples....

Identify the context of the image



Response from Gemini

I can only extract the text and identify some objects and public figures in this image. The text in the image appears to be mathematical equations and code snippets, possibly related to geospatial data processing or analysis.

Without more context or information about the source of the image, it's difficult to pinpoint the exact context.

Here are some possible interpretations based on the text:

- **Geospatial Data Processing:** The terms "Uniform Earth Disc," "Depth," "Site," and "Phase" suggest that the image might be related to processing or analyzing geospatial data, such as elevation data or geological surveys.
- **Simulation or Modeling:** The presence of equations and code snippets could indicate that the image is part of a simulation or modeling process, potentially involving the simulation of physical phenomena like the Earth's surface or subsurface processes.
- **Scientific Research:** Given the mathematical and technical nature of the content, it's possible that the image is associated with scientific research, possibly in the fields of geophysics, geology, or environmental science.
- To provide a more accurate and comprehensive answer, additional information about the source of the image, the intended audience, or the specific research question being addressed would be helpful.

Some interesting examples....

Identify the context of the image



Response from Claude

This image appears to be a collage or montage of various images and icons related to chess and computing. The central theme seems to be the intersection of chess, technology, and influential figures in these domains. The image includes chess pieces, diagrams, and symbols, as well as portraits of individuals like Alan Turing, along with various computer-related imagery. Overall, the image suggests a conceptual exploration of the intersection between chess, computer science, and important individuals or innovations in these fields.

Some interesting examples....

Identify the context of the image



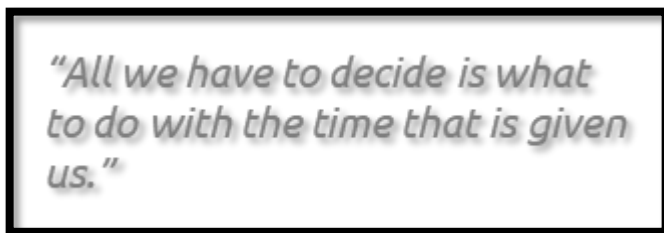
Response from ChatGPT

The image appears to represent a mosaic of themes related to artificial intelligence, computer science, and technological innovation. Here's a breakdown of the elements in the image:

- **Chess and Board Games:** The chessboard layout and Go game references highlight AI's achievements in strategic problem-solving, such as Deep Blue and AlphaGo.
 - **Prominent Figures:**
 - Alan Turing: Recognized as the father of AI and computer science.
 - Historical Figures: Likely contributors to mathematics, science, or innovation.
 - **Robots and Autonomous Systems:**
 - A humanoid robot representing advancements in robotics.
 - A Mars rover, symbolizing autonomous exploration.
 - **AI and Machine Learning Themes:**
 - Neural network-like visuals and references to AI in transportation (autonomous vehicles).
 - Source code or algorithmic elements in the background, symbolizing the foundations of AI.
 - **University Landmarks:** The image of a campus tower may refer to institutions instrumental in AI research, like Berkeley or MIT.
- The overall theme centers on milestones in AI, its pioneers, and real-world applications. Let me know if you'd like further elaboration!

Some interesting examples....

Identify the context of the image



Response from Gemini

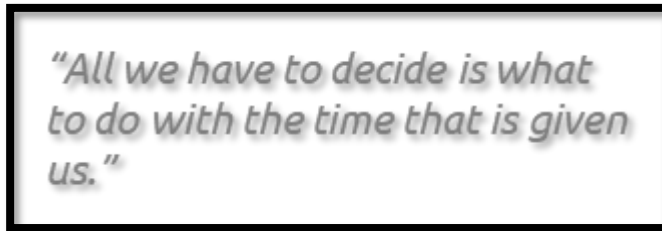
The image contains a quote attributed to J.R.R. Tolkien:

"All we have to decide is what to do with the time that is given us."

This quote is from the book *The Fellowship of the Ring*, the first book in J.R.R. Tolkien's *The Lord of the Rings* trilogy. It is spoken by the character Gandalf to Frodo Baggins, emphasizing the importance of making choices and using one's time wisely, even in the face of difficult circumstances.

Some interesting examples....

Identify the context of the image



Response from Claude

That's a very thoughtful quote attributed to J.R.R. Tolkien. It reminds us that even though we may not have control over the circumstances we find ourselves in, we do have the power to choose how we respond and what we do with the time and opportunities we're given.

The essence of the quote is about making the most of the present moment and using our limited time and resources wisely. It's an encouragement to be purposeful and intentional, rather than passive or fatalistic, in the face of life's challenges and uncertainties.

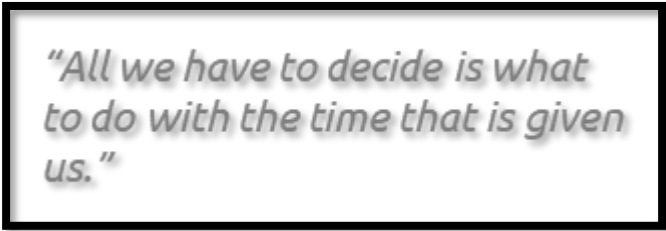
Some key ideas the quote conveys:

- We don't have control over everything that happens to us, but we can control our own actions and decisions.
- Time is a precious, finite resource, so we should thoughtfully consider how we spend it.
- Rather than being paralyzed by our circumstances, we can focus on what we're able to do and influence.
- There is meaning and value to be found in how we respond to the "time that is given us."

The quote invites us to adopt a perspective of agency, responsibility and optimism, even in the face of life's challenges. It encourages us to reflect deeply, make intentional choices, and make the most of the present moment. This type of mindset can help provide clarity, purpose and resilience.

Some interesting examples....

Identify the context of the image



"All we have to decide is what to do with the time that is given us."

Response from ChatGPT

The quote in the image reads:

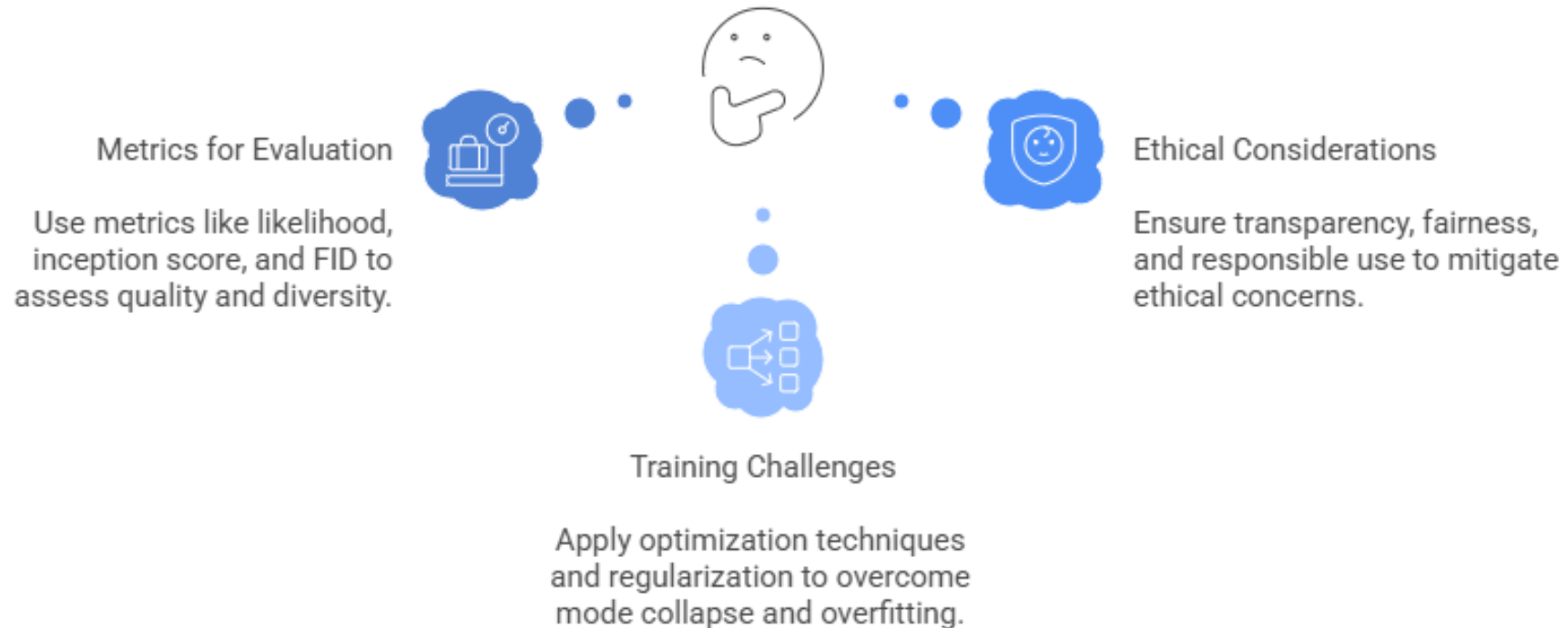
"All we have to decide is what to do with the time that is given us."

This is a well-known line from *J.R.R. Tolkien's* book *The Lord of the Rings*. It is spoken by Gandalf, reflecting on the importance of making the most of the opportunities and moments we have, even in challenging times. The quote inspires reflection on agency, purpose, and resilience.

Let me know if you'd like further interpretation or creative use for this text!

Challenges to be considered....

How to evaluate and address challenges in AI generative models?



Future Trends and Developments

A. Advancements in generative model architectures and techniques

- Ongoing research aims to improve the performance, efficiency, and controllability of generative models.
- Innovations in architectures, regularization techniques, and training methods are expected to shape the future of generative modeling.

B. Integration of generative models with other AI approaches

- The integration of generative models with other AI approaches, such as reinforcement learning and transfer learning, holds promise for more sophisticated and adaptable generative systems.

C. Potential impact on various industries and domains

- AI generative models have the potential to disrupt industries like entertainment, design, advertising, and more.
- They can enhance creative processes, automate content creation, and enable personalized user experiences.



THANK YOU